

Case Study

Maa Laxmi

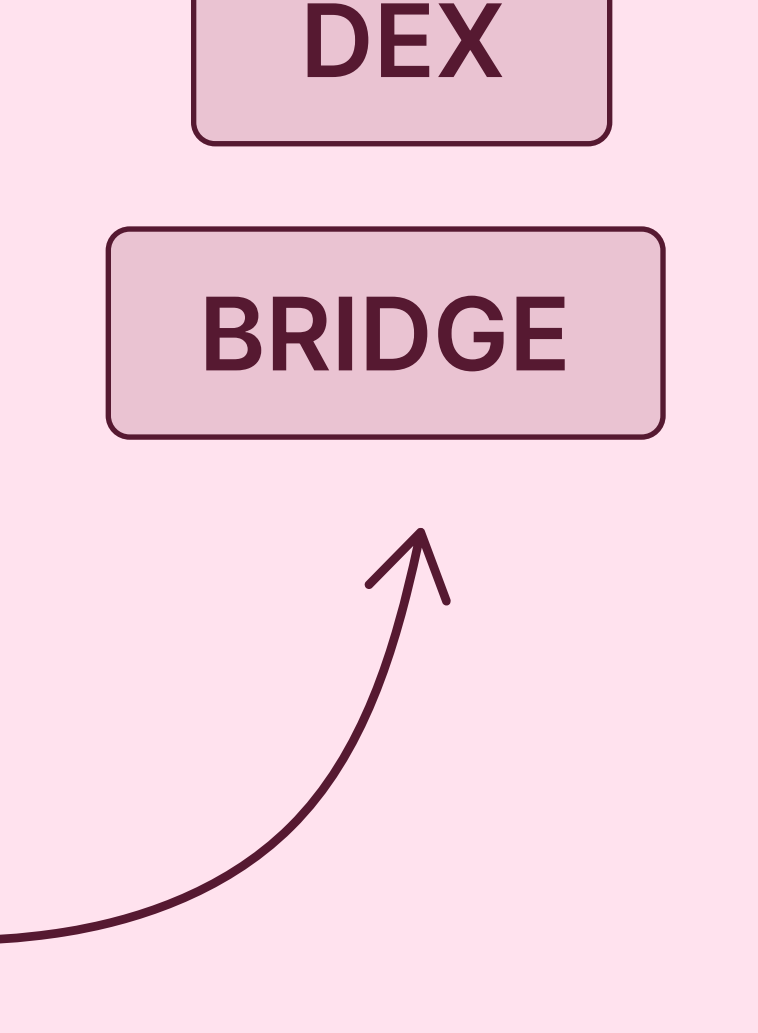
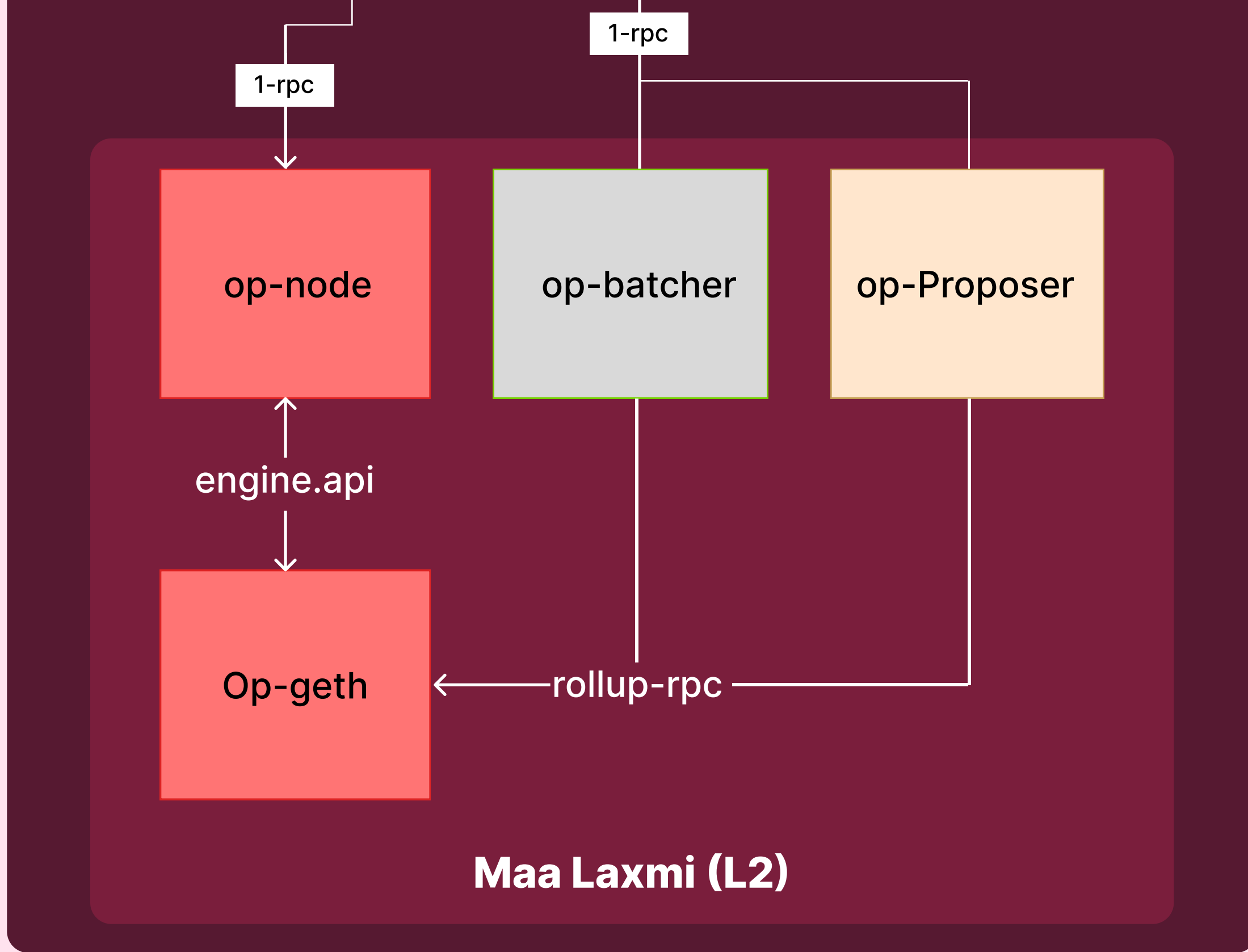
L2 blockchain based on OP-Stack



Project Summary

The Maa Laxmi project aims to enhance an existing blockchain ecosystem by configuring and deploying a new L2 blockchain using OP-Stack technology. Key steps include integrating the Laxmi coin for blockchain operations and implementing fee payments in this native currency. The project also involves developing and deploying a blockchain explorer for real-time tracking and monitoring.

CONCEPT VISUALISATION



Services Used

- AWS / MySQL Server/.Net
- Micro Services Architecture And Development
- CI/CD Implementation And Management



Technology Stacks

- OP-Stack
- Blockscout
- GETH
- Ganache
- Hardhat
- Solidity
- Next.js
- Web3.js
- Bash Scripts
- ElixirJS
- NodeJS



Objective

The Maa Laxmi project aims to configure and deploy a new blockchain using open-source technologies, update branding to match Laxmi Chain, and provide extensive documentation and deployment scripts. Key objectives include configuring and deploying the blockchain, updating branding elements, and setting up a Git repository for the client. The primary outcomes are a fully operational Laxmi Chain blockchain, branded frontend and backend, comprehensive deployment documentation, and automated deployment scripts. These deliverables enhance system functionality, user experience, and ease of maintenance, providing greater transparency, security, and efficiency for users and stakeholders, while improving interoperability and supporting seamless token trading and blockchain exploration.

Key Features

- OnChain Definition of L2 Chain**
Record critical information defining the L2 chain directly on the blockchain, including the generation of a unique chain ID and configuration of key parameters such as the block gas limit.
- Deployment of Laxmi Blockchain Foundation Node and Testnet Explorer**
Establish the foundation node for the L2 blockchain on the OP Stack, ensuring seamless integration with a testnet explorer for monitoring and exploration.

- Smart Contract Factory Implementation**
Develop a smart contract factory to streamline the deployment of configuration and necessary contracts for each L2 chain. Utilise CREATE2 for deterministic contract addresses, enabling easy interaction without explicit deployment.
- Local Computation for OP Chain Sync**
Ensure the OP Chain state is locally computed, eliminating the need for a proof system during chain derivation. Achieve a permission less and secure determination of the OP Chain state without relying on a proof system.

- Withdrawal Mechanism Enhancements**
Introduce withdrawal claims, enabling anyone to submit a withdrawal proposal. Remove the interval for proposal submission, allowing users to make withdrawal claims as needed.



Maalaxmi Benefits

- Lower Transaction Fees**
By offloading transactions from the main L1 chain, L2 chains reduce the demand on L1, resulting in lower gas fees for users.
- Increased Throughput**
L2 solutions like OP Stack can handle a significantly higher number of transactions per second (TPS) compared to L1 chains, alleviating congestion and improving performance.
- Long-Term Viability**
By alleviating pressure on L1 chains, L2 solutions contribute to the long-term sustainability and scalability of blockchain networks, ensuring they can handle growing demand without compromising on performance or security.



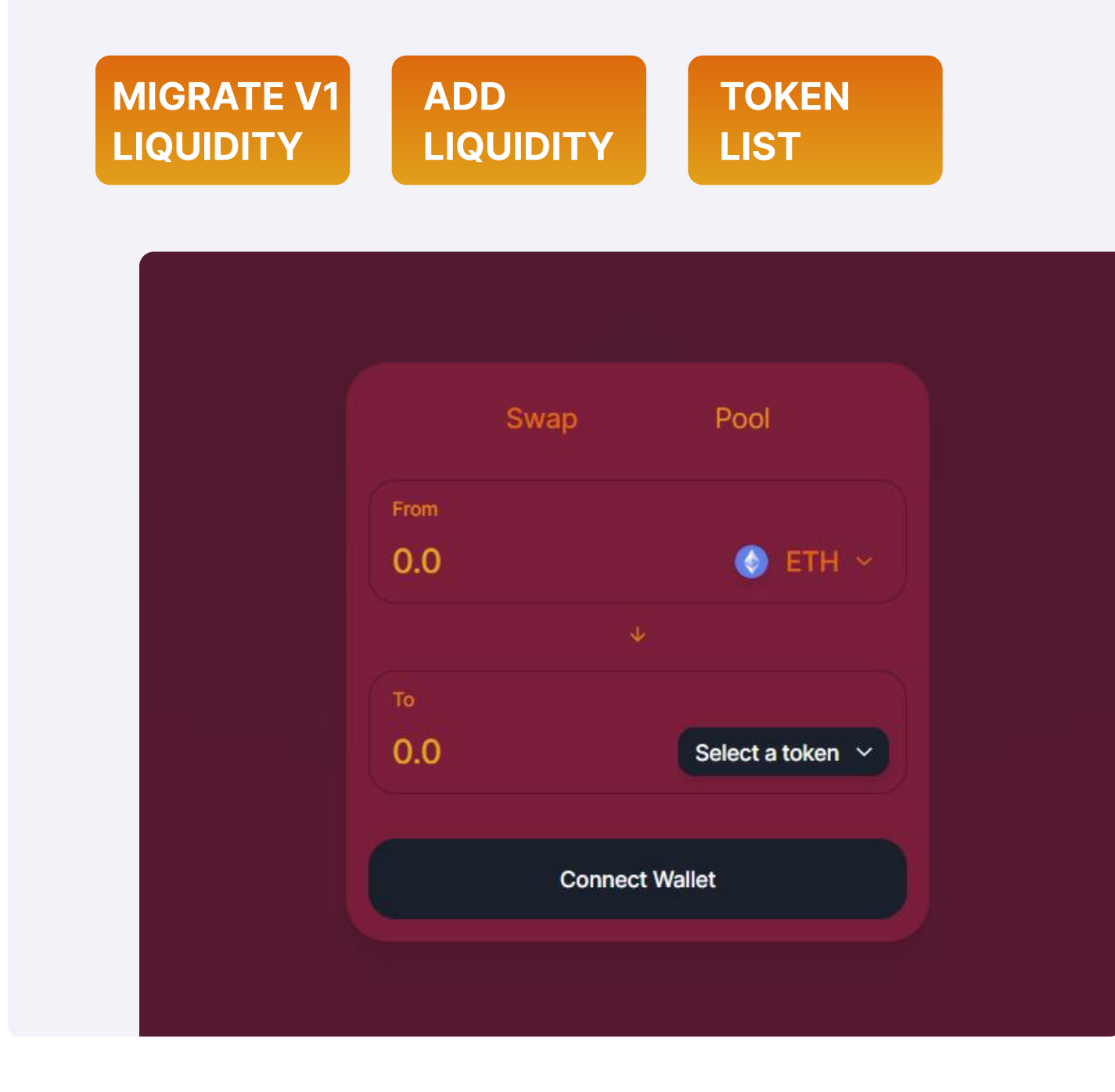
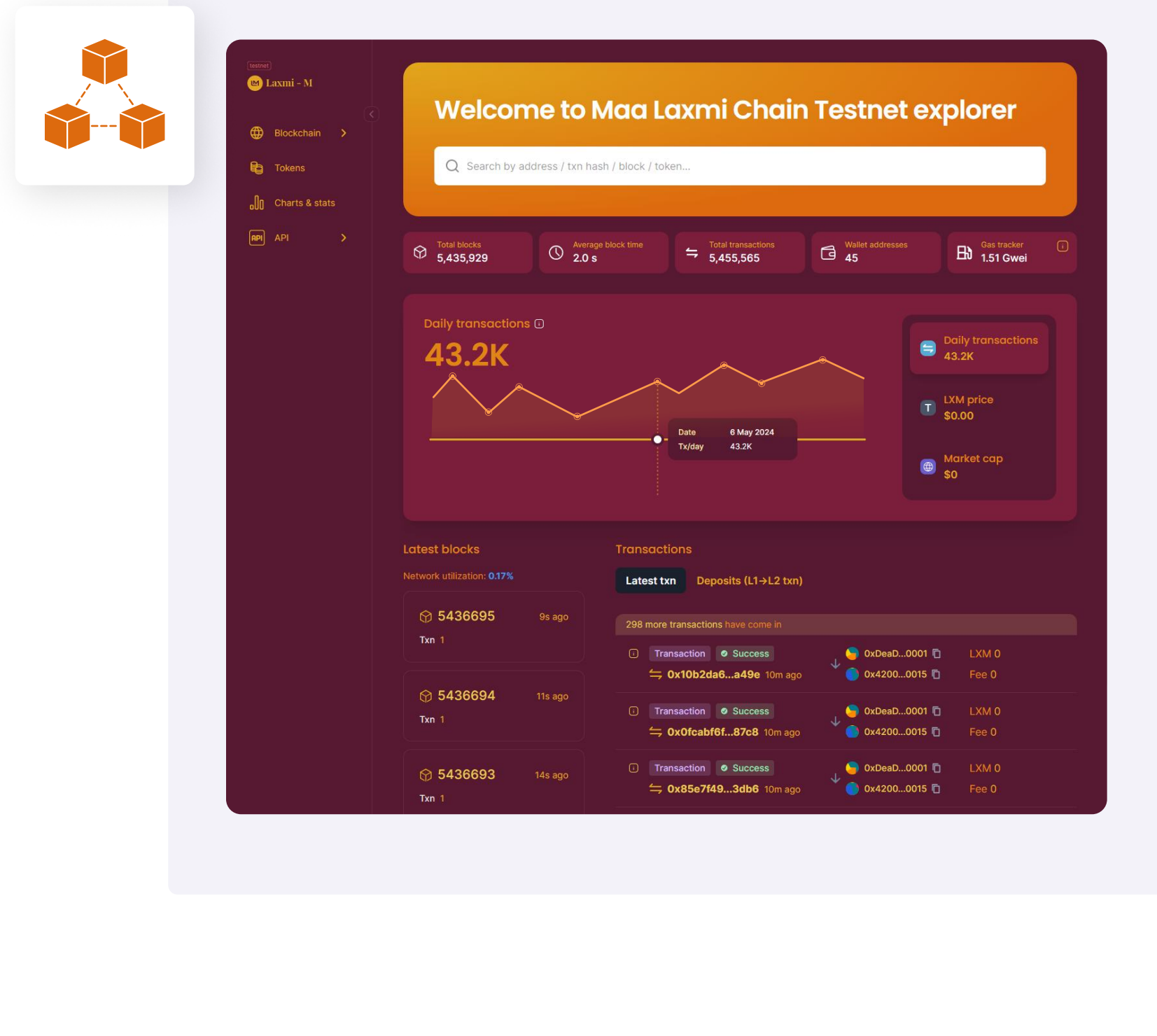
Challenges

- ERC-20 Implementation**
LAXMI is an ERC-20 token, new smart contracts need to be deployed or existing ones updated to support LAXMI transactions.
- Deposit and Withdrawal**
Update the deposit and withdrawal processes to handle LAXMI instead of ETH.
- Bridging Contracts**
Update or deploy new bridging contracts to facilitate the transfer of LAXMI between L1 and L2.
- Cross-Layer Communication**
Ensuring that the state of the L2 chain is consistently and accurately reflected on the L1 chain requires sophisticated mechanisms for state synchronization. Any discrepancy can lead to inconsistencies and potential security vulnerabilities.
- Cross-Layer Transfers**
Implement mechanisms to facilitate seamless transfer of LAXMI between L1 and L2.
- Contract Rewrites**
Rewrite L2 smart contracts to accept and operate with LAXMI.
- Upgradability**
Upgrading the L2 protocol without disrupting the existing system and maintaining backward compatibility is complex.
- Transaction Finality**
Transactions need to be finalized on L2 before being committed to L1. Achieving fast finality while maintaining security and decentralization is a technical challenge.



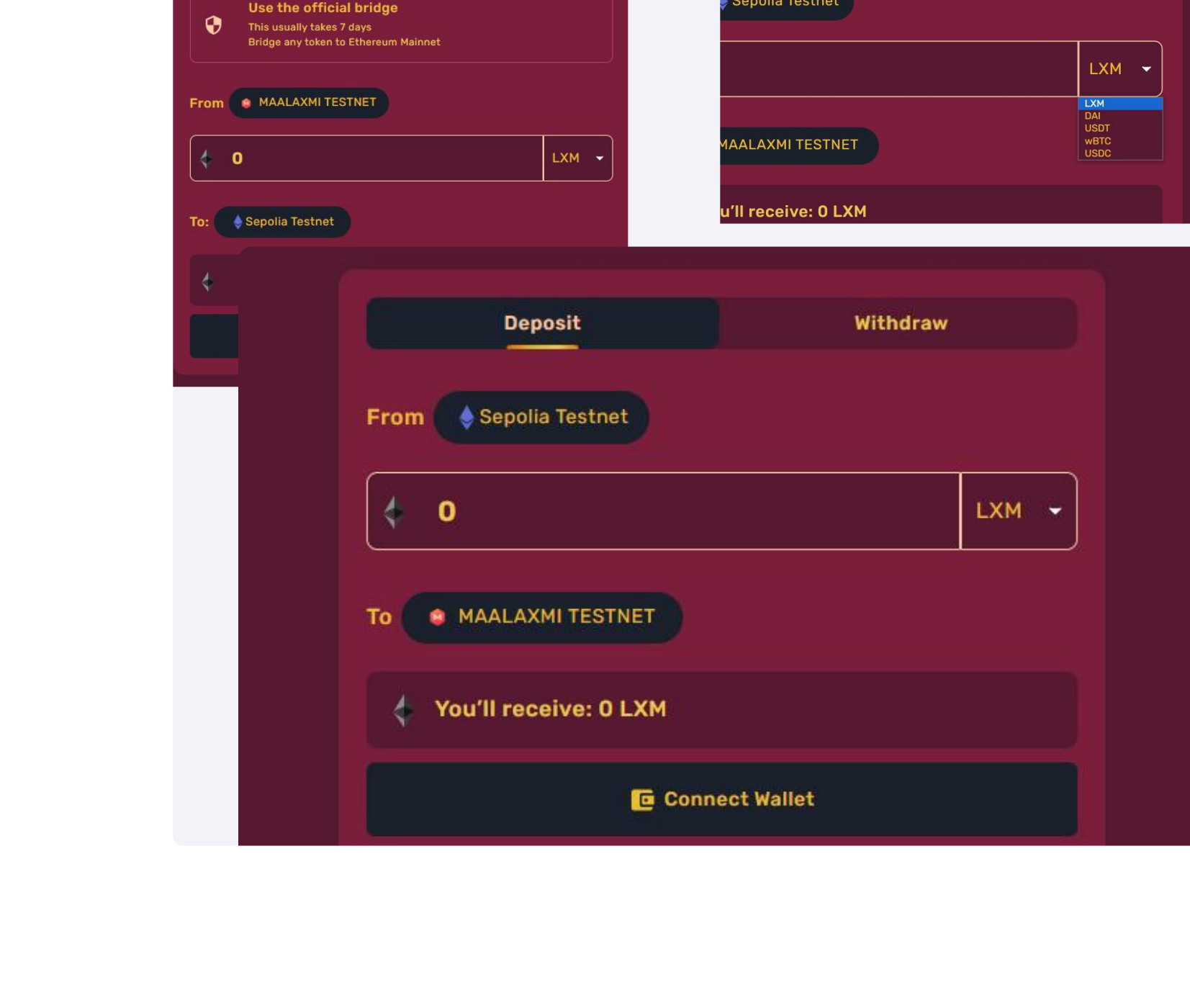
Our Solutions

- RPC NODE INFORMATION**
- Custom Blockchain Explorer**
 - Tailored to provide detailed insights into transactions, blocks, and overall network activity on the Laxmi L2 blockchain.
 - Enhanced with features for improved user experience and detailed analytics.

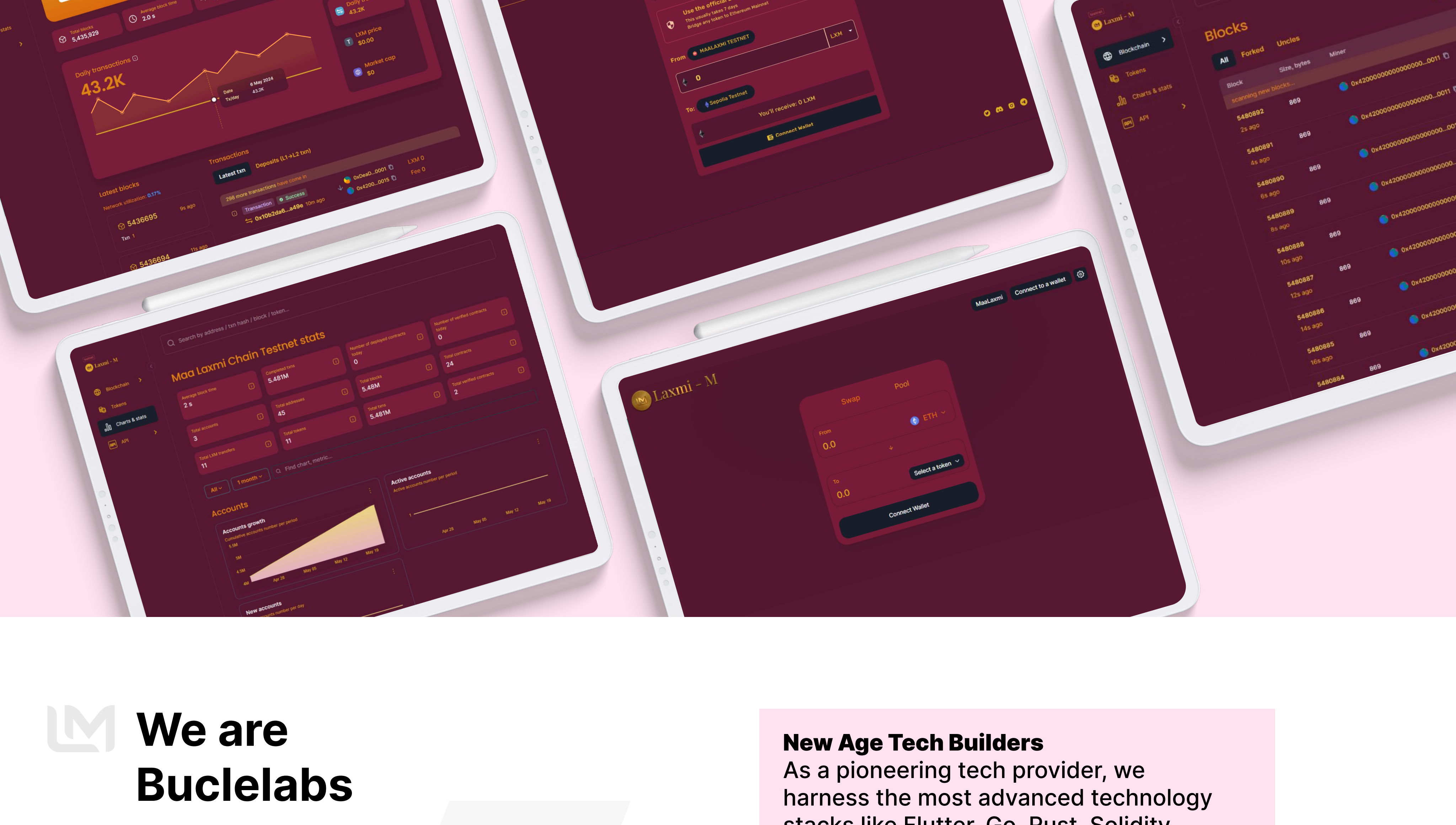


- INDEPENDENT EXCHANGE**
- Decentralized Exchange (Dex)**
 - A bespoke platform for trading digital assets directly on the Laxmi L2 blockchain.
 - Enhanced liquidity, custom trading pairs, and user-friendly interfaces to cater to the specific needs of users.

- L1- L2 TELEPORTER**
- Customized Bridge**
 - Designed to ensure seamless interoperability between the Laxmi L2 blockchain and other blockchain networks.
 - Facilitates secure and efficient asset transfers with custom security protocols.



Design Guidelines



We are Buclelabs

BucleLabs deliver unmatched software development services that empower businesses to thrive in the digital era. From established enterprises to emerging startups, we provide top-tier solutions crafted to their needs

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New Age Tech Builders

As a pioneering tech provider, we harness the most advanced technology stacks like Flutter, Go, Rust, Solidity and more.

Security-First Approach

Our commitment to security is ingrained in our "secure-by-design" philosophy.

Culture of Innovation

Our commitment to security is ingrained in our "secure-by-design" philosophy.

Innovative and Tech-Savvy

Confident and Charismatic

Solution Oriented

Entrepreneurial Spirit

Thankyou